Qingyu Li

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EDUCATION

04/2019 – 11/2022 **Doctor of Engineering** in Geoscience, *Technical University of Munich*

09/2015 – 06/2019 **Master of Engineering** in Photogrammetry and Remote Sensing, *Wuhan University*

10/2016 – 11/2018 **Master of Science** in Earth Oriented Space Science and Technology, *Technical University of Munich*

09/2011 – 06/2015 **Bachelor of Engineering** in Remote Sensing Science and Technology, *Wuhan University*

PROFESSIONAL APPOINTMENTS

01/2025 – Now **Assistant Professor**, *The Chinese University of Hong Kong, Shenzhen*

11/2022 – 12/2024 **Postdoctoral Researcher**, *Technical University of Munich*

04/2019 – 11/2022 **Research Associate**, *Technical University of Munich & German Aerospace Center*

TEACHING EXPERIENCE

01/2025 – Now **Instructor of Course:** MCE5908 Big Data Systems and Information Processing;
CIE6006/MCE5918 Data Analytics
ECE3001 Signals and Systems

SERVICE & LEADERSHIP

- **IEEE GRSS Shenzhen Chapter chair**
- **Session chair**

(1) 2022 IEEE International Geoscience and Remote Sensing Symposium – WE3.O1: Image Segmentation and Mapping;
(2) 2021 IEEE International Geoscience and Remote Sensing Symposium – WE1.O.3: Advanced Segmentation and Land Cover Methods for Optical Data

- **Reviewer** for Scholarly Journals

(1) IEEE Transactions on Geoscience and Remote Sensing; (2) International Journal of Applied Earth Observation and Geoinformation; (3) ISPRS Journal of Photogrammetry and Remote Sensing; (4) Remote Sensing of Environment; (5) Remote Sensing; (6) IEEE Geoscience and Remote Sensing Letters; (7) IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing; (8) IEEE Transactions on Industrial Informatics; (9) Geo-Spatial Information Science; (10) Earth System Science Data; (11) Applied Energy; (12) Sustainable Cities and Society; (13) Building and Environment

SELECTED AWARDS

10/2020 **Geodesy Award** of the German Association for Geodesy, Geoinformation, and Land Management

TALKS

08/2025 "Identification of Roof Materials in Urban Hot Spots from Remote Sensing Imagery Using Deep Learning Methods." Oral presentation at the 2025 IEEE International Geoscience and Remote Sensing Symposium.

04/2025 " Evaluating roof greening Feasibility across cities using remote sensing and deep learning." Oral presentation at the 2025 International Symposium on Digital Earth.

05/2023 "Semi-supervised segmentation of individual buildings from SAR imagery." Oral presentation at the 2023 Joint Urban Remote Sensing Event.

07/2022 "Feature and Output Consistency Training for Semi-supervised Building Footprint Generation." Oral presentation at the 2022 IEEE International Geoscience and Remote Sensing Symposium.

07/2021	"End-to-End Semantic Segmentation and Boundary Regularization of Buildings from Satellite Imagery." Oral presentation at the 2021 IEEE International Geoscience and Remote Sensing Symposium.
10/2020	"Mapping the land cover of Africa at 10 m resolution from multi-source remote sensing data with Google Earth Engine." Oral presentation at the 2020 Phi-week, European Space Agency
09/2020	"Instance Segmentation of Buildings Using Keypoints." Oral presentation at the 2020 IEEE International Geoscience and Remote Sensing Symposium.
09/2020	"Detection of Undocumented Buildings using Convolutional Neural Network and Official Geodata." Oral presentation at the 2020 XXIVth ISPRS Congress.

PUBLICATIONS

Book chapters

[1] Roschlaub, R., Glock, C., Möst, K., **Li, Q.**, Auer, S., & Zhu, X. X. (2022). "Mit Deep Learning und amtlichen Daten zur landesweiten Detektion von Gebäuden und Gebäudeveränderungen." in Künstliche Intelligenz in Geodäsie und Geoinformatik - Potenziale und Best-Practice-Beispiele, edited by Grunau, Wilfried. Germany: Wichmann Verlag.

Journal papers

[2] Yao, J., Krapf, S., & **Li, Q***. (2026). Multimodal deep learning with aerial imagery and nDSM for true-slope-aware rooftop photovoltaic potential estimation. ISPRS Journal of Photogrammetry and Remote Sensing, 238, 693-711.

[3] Tang, Z., Mu, Y., Gonzalez-Alonso, M., & **Li, Q***. (2026). A satellite-based assessment of alder pollen exposure across 51 cities in Bavaria, Germany. Urban Forestry & Urban Greening, 129548.

[4] Zhu, X. X., **Li, Q.**, Shi, Y., Wang, Y., Stewart, A. J., Prexl, J., & Zhang, F. (2026). GlobalBuildingMap—Unveiling the mystery of global buildings. Scientific Data, 13, 71.

[5] **Li, Q.**, Xu, G., & Gu, Z. (2025). Deep learning and remote sensing for scalable building age prediction in urban energy modeling. Energy and Buildings, 116303.

[6] **Li, Q.**, Mou, L., Shi, Y., & Zhu, X. X. (2025). BANet: A bilateral attention network for extracting changed buildings between remote sensing imagery and cadastral maps. International Journal of Applied Earth Observation and Geoinformation, 139, 104486.

[7] **Li, Q.**, Taubenböck, H., & Zhu, X. X. (2025). Identification of the potential for roof greening using remote sensing and deep learning. Cities, 159, 105782.

[8] **Li, Q.**, Krapf, S., Mou, L., Shi, Y., & Zhu, X. X. (2024). Deep learning-based framework for city-scale rooftop solar potential estimation by considering roof superstructures. Applied Energy, 374, 123839.

[9] **Li, Q.**, Xu, G., & Gu, Z. (2024). A novel framework for multi-city building energy simulation: Coupling urban microclimate and energy dynamics at high spatiotemporal resolutions. Sustainable Cities and Society, 113, 105718.

[10] **Li, Q.**, Mou, L., Sun, Y., Hua, Y., Shi, Y., & Zhu, X. X. (2024). A Review of Building Extraction from Remote Sensing Imagery: Geometrical Structures and Semantic Attributes. IEEE Transactions on Geoscience and Remote Sensing, 62,1-15.

[11] Liu, C., Albrecht, C. M., Wang, Y., **Li, Q.**, & Zhu, X. X. (2024). AIO2: Online Correction of Object Labels for Deep Learning with Incomplete Annotation in Remote Sensing Image Segmentation. IEEE Transactions on Geoscience and Remote Sensing.

[12] **Li, Q.**, Mou, L., Hua, Y., Shi, Y., Chen, S., Sun, Y., & Zhu, X. X. (2023). 3DCentripetalNet: Building height retrieval from monocular remote sensing imagery. International Journal of Applied Earth Observation and Geoinformation, 120, 103311.

[13] **Li, Q.**, Krapf, S., Shi, Y., & Zhu, X. X. (2023). SolarNet: A convolutional neural network-based framework for rooftop solar potential estimation from aerial imagery. International Journal of Applied Earth Observation and Geoinformation, 116, 103098.

[14] **Li, Q.**, Taubenböck, H., Shi, Y., Auer, S., Roschlaub, R., Glock, C., Kruspe, A., & Zhu, X. X. (2022). Identification of undocumented buildings in cadastral data using remote sensing: Construction period, morphology, and landscape. International Journal of Applied Earth Observation and Geoinformation, 112, 102909.

[15] Roschlaub, R., Glock, C., Möst, K., Hümmer, F., **Li, Q.**, Auer, S., Kruspe, A., & Zhu, X. X. (2022). Implementierung einer KI-Infrastruktur zur automatisierten Erkennung von landesweiten Gebäudeveränderungen aus Luftbildern. ZfV-Zeitschrift für Geodäsie, Geoinformation und Landmanagement, (zfv 3/2022).

[16] **Li, Q.**, Shi, Y., & Zhu, X. X. (2022). Semi-supervised building footprint generation with feature and output consistency training. IEEE Transactions on Geoscience and Remote Sensing, 60, 1-17.

[17] **Li, Q.**, Zorzi, S., Shi, Y., Fraundorfer, F., & Zhu, X. X. (2022). RegGAN: An end-to-end network for building footprint generation with boundary regularization. Remote Sensing, 14(8), 1835.

[18] **Li, Q.**, Mou, L., Hua, Y., Shi, Y., & Zhu, X. X. (2022). CrossGeoNet: A framework for building footprint generation of label-scarce geographical regions. International Journal of Applied Earth Observation and Geoinformation, 111, 102824.
[19] **Li, Q.**, Mou, L., Hua, Y., Shi, Y., & Zhu, X. X. (2021). Building footprint generation through convolutional neural networks with attraction field representation. IEEE Transactions on Geoscience and Remote Sensing, 60, 1-17.
[20] **Li, Q.**, Shi, Y., Auer, S., Roschlaub, R., Möst, K., Schmitt, M., Glock, C., & Zhu, X. X. (2020). Detection of undocumented building constructions from official geodata using a convolutional neural network. Remote Sensing, 12(21), 3537.
[21] **Li, Q.**, Shi, Y., Huang, X., & Zhu, X. X. (2020). Building footprint generation by integrating convolution neural network with feature pairwise conditional random field (FPCRF). IEEE Transactions on Geoscience and Remote Sensing, 58(11), 7502-7519.
[22] **Li, Q.**, Qiu, C., Ma, L., Schmitt, M., & Zhu, X. X. (2020). Mapping the land cover of Africa at 10 m resolution from multi-source remote sensing data with Google Earth Engine. Remote Sensing, 12(4), 602.
[23] Roschlaub, R., **Li, Q.**, Auer, S., Möst, K., Glock, C., Schmitt, M., Shi, Y & Zhu, X. X. (2020). KI-basierte Detektion von Gebäuden mittels Deep Learning und amtlichen Geodaten zur Baufallerkundung. ZFV-Zeitschrift für Geodasie, Geoinformation und Landmanagement, (3), 180-189.
[24] Shi, Y., **Li, Q.**, & Zhu, X. X. (2020). Building segmentation through a gated graph convolutional neural network with deep structured feature embedding. ISPRS Journal of Photogrammetry and Remote Sensing, 159, 184-197.
[25] Shi, Y., **Li, Q.**, & Zhu, X. X. (2018). Building footprint generation using improved generative adversarial networks. IEEE Geoscience and Remote Sensing Letters, 16(4), 603-607.
[26] **Li, Q.**, Huang, X., Wen, D., & Liu, H. (2017). Integrating multiple textural features for remote sensing image change detection. Photogrammetric Engineering & Remote Sensing, 83(2), 109-121.
[27] Huang, X., **Li, Q.**, Liu, H., & Li, J. (2016). Assessing and improving the accuracy of GlobeLand30 data for urban area delineation by combining multisource remote sensing data. IEEE Geoscience and Remote Sensing Letters, 13(12), 1860-1864.

Conference papers

[28] **Li, Q.**, & Yang, X. (2025). Identification of Roof Materials in Urban Hot Spots from Remote Sensing Imagery Using Deep Learning Methods. In IGARSS 2025-2025 IEEE International Geoscience and Remote Sensing Symposium. IEEE.
[29] Lin, W., Zhu, J., Hua, Y., Li, Q., Mou, L., & Zhu, X. X. (2024). Towards Sustainable Urban Energy: A Robust Deep Learning Framework for Solar Potential Estimation. The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, 48, 371-378.
[30] **Li, Q.**, Krapf, S., Mou, L., Shi, Y., & Zhu, X. X. (2023). Roof superstructure detection from aerial imagery. In IGARSS 2023-2023 IEEE International Geoscience and Remote Sensing Symposium. IEEE.
[31] **Li, Q.**, Sun, Y., Mou, L., Shi, Y., & Zhu, X. X. (2023). Semi-supervised segmentation of individual buildings from SAR imagery. In 2023 Joint Urban Remote Sensing Event (JURSE). IEEE.
[32] **Li, Q.**, Shi, Y., & Zhu, X. X. (2022). Feature and output consistency training for semi-supervised building footprint generation. In IGARSS 2022-2022 IEEE International Geoscience and Remote Sensing Symposium. IEEE.
[33] **Li, Q.**, Zorzi, S., Shi, Y., Fraundorfer, F., & Zhu, X. X. (2021). End-to-end semantic segmentation and boundary regularization of buildings from satellite imagery. In IGARSS 2021-2021 IEEE International Geoscience and Remote Sensing Symposium. IEEE.
[34] Chen, S., Mou, L., **Li, Q.**, Sun, Y., & Zhu, X. X. (2021). Mask-height R-CNN: An end-to-end network for 3D building reconstruction from monocular remote sensing imagery. In IGARSS 2021-2021 IEEE International Geoscience and Remote Sensing Symposium. IEEE.
[35] **Li, Q.**, Mou, L., Hua, Y., Sun, Y., Jin, P., Shi, Y., & Zhu, X. X. (2020). Instance segmentation of buildings using keypoints. In IGARSS 2020-2020 IEEE International Geoscience and Remote Sensing Symposium. IEEE.
[36] **Li, Q.**, Shi, Y., Auer, S., Roschlaub, R., Möst, K., Schmitt, M., & Zhu, X. X. (2020). Detection of Undocumented Buildings Using Convolutional Neural Network and Official Geodata. ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences, 2, 517-524.
[37] Shi, Y., **Li, Q.**, & Zhu, X. X. (2020). Building extraction by gated graph convolutional neural network with deep structured feature embedding. In IGARSS 2020-2020 IEEE International Geoscience and Remote Sensing Symposium. IEEE.
[38] Shi, Y., **Li, Q.**, & Zhu, X. X. (2019). Building footprint extraction with graph convolutional network. In IGARSS 2019-2019 IEEE International Geoscience and Remote Sensing Symposium. IEEE.
[39] Shi, Y., **Li, Q.**, & Zhu, X. X. (2019). BFGAN–building footprint extraction from satellite images. In 2019 Joint Urban Remote Sensing Event (JURSE). IEEE.